Practice: Confidence Intervals for Means

Module 4 | Lesson 3 | Quantify uncertainty about a mean

Use the lesson for the checked case introductions and source limits. Submit a short Quarto report with calculations, units, assumptions, and interpretation.

1. Reproduce the 95% Lloyd's interval and calculate a 99% interval using the same summaries. Compare widths.
2. Explain why neither interval describes the range of individual customer spending.
3. Reproduce the 95% household-balance t interval. Calculate a 90% interval; identify df, estimated SE, and margin.
4. Explain why using z merely because n = 70 is inappropriate when population SD is unknown.
5. Explain 95% confidence without assigning 95% probability to the fixed population mean. Name two assumptions that summaries alone cannot verify.